

CSE490 (D)

INTRODUCTION TO FOOTBALL DATA AND ANALYTICS

IN ASSOCIATION WITH BANGLADESH FOOTBALL FEDERATION

SPRING 2026



- Professional Certificate from BFF
- Internship Opportunity at BFF Data Science Department after successful completion
- Learning from the industry experts of Football Data Science
- Opportunity to co-author research papers for Sports Data Conferences

CSE490 (D)

Introduction to Football Data & Analytics

Lecture-9

Pitch Control

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Context

- ❖ Only 1 player has the ball
- ❖ What others do when they don't have the ball?
- ❖ Self read: Soccermetrics book– Chapter 2 (included in syllabus)

Tracking Data

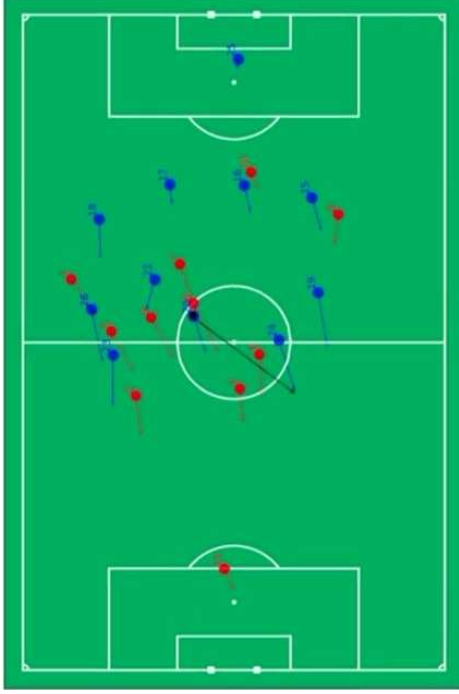
- Event data only focuses on on-ball action
- **Optical Tracking Data:**
 - 2D positions of players in a visible area (depends on camera position)
- **GPS Tracking + Vital Data:**
 - 2D positions and movement data along with vital data (for example, heart rate).
 - Have to wear vest/sensor.
 - Only own team data available.



Optical Tracking Data

- 2D positions of players in a visible area (depends on camera position)
- We can represent each player with vector. Length corresponds to the speed and arrow corresponds to the direction of running.
- Can be merged with event data (black line indicates pass here)

Can you guess the type of play?



Optical Tracking Data

Limitations:

- Does not account for body position. (Backtracking)
- Sometimes error prone (ID switches).
- Ball usually tracked in 2d. So some strange trajectories.
- Need to link it to event data.
- Lack of public data.

Voronoi Diagram

- Way of dividing space into regions based on proximity to a set of points (called seeds or sites).
- Each region, called a Voronoi cell, contains all the locations closer to its seed than to any other seed.

Voronoi Diagram

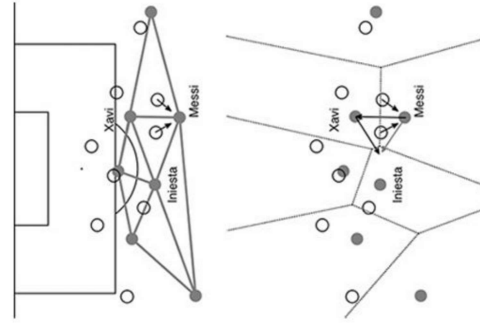


Figure 2.5 Barcelona's positions outside the Panathinaikos penalty area five seconds before Messi's goal. Barcelona attacking bottom to top. Barcelona players' positions are marked with grey circles, Panathinaikos players' positions are open circles. Arrows indicate that the two defenders are running towards Messi. The passing triangles are given by the direct links between the players (top panel); the player zones are given by the dotted lines (bottom panel).

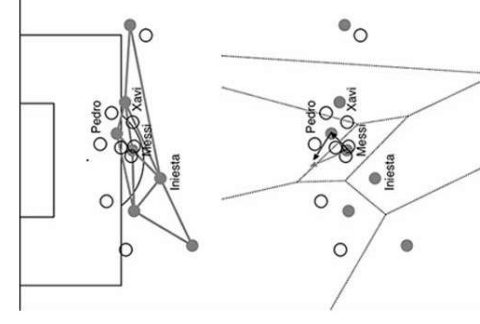


Figure 2.6 Barcelona's positions outside the Panathinaikos penalty area three seconds before Messi's goal. See Figure 2.5 for explanation of the symbols.

As a defender, you don't wanna be on the boundary!!

Passing Triangles

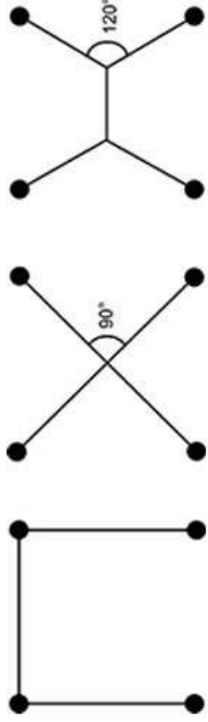
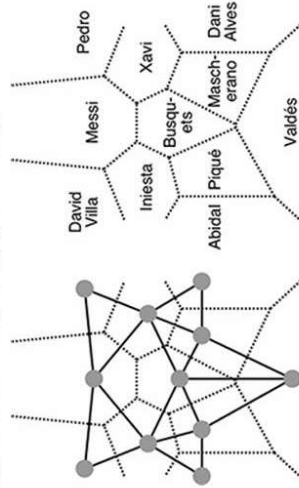


Figure 2.2 Three potential solutions for linking up four suburbs (circles) with the least possible train track (solid lines).



Triangles solve many connecting problems including football!

Figure 2.4 The Barcelona 2010/11 network and zones. The formation network (solid lines) together with the zones (dotted lines) for each player (left); typical

Rules of Motion

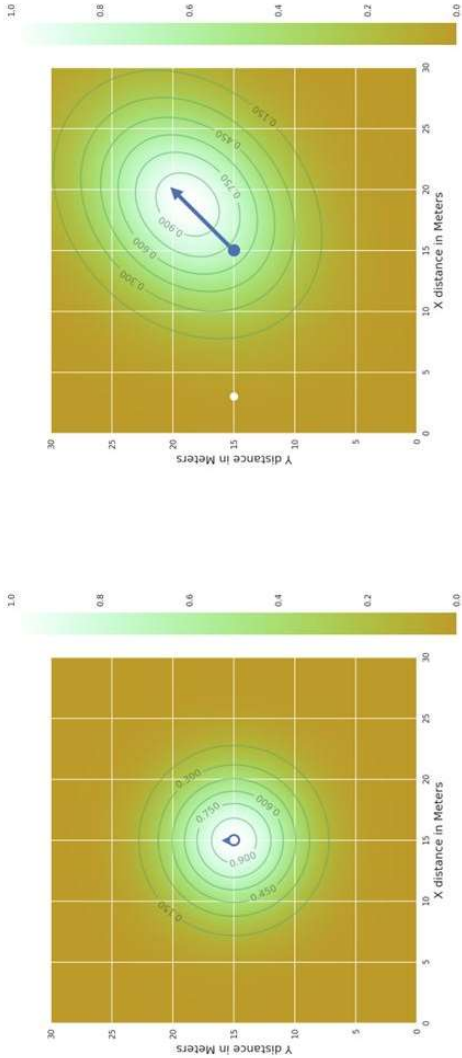
[Million Sardines Create Living Tornado Around Freedivers in Bait Ball](#)

- ❖ Fish create complex group patterns **without understanding them**, patterns emerge from simple behavior.
- ❖ Each fish only follows **nearby neighbors**, adjusting speed and direction to stay with the group.
- ❖ **Simple local rules** → **complex global behavior** (no central control needed).
- ❖ Can you relate this with players' movement? Think about **offside-trap**.



Not a good trap!

Player Influence Area



(a) Player influence area for player in possession of the ball and no speed (lower than walking speed)
(b) Player influence area for player 15 meters away from the ball, running at 6.36 m/s in a 45 degrees angle

Figure 2: Two situations representing the player influence area

Pitch Control Surface

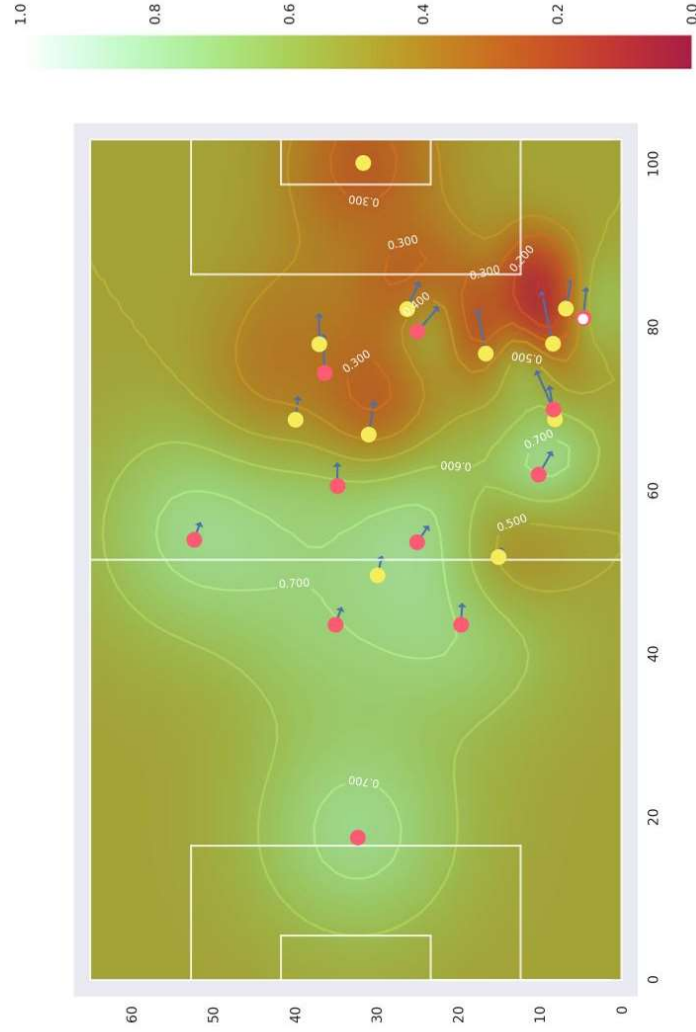


Figure 3: Pitch control surface indicating the degree of control for team in red. Arrows show the magnitude and direction of speed, and contour lines allow to visualize the surface geometry. Numbers in white indicate the pitch control value at their drawing location. Axis dimensions are in meters/

Building a Pitch Control Model

Pitch control at a given location is the probability that a player (or team) will gain control of the ball if it moves directly to that location.

For a given location on a pitch:

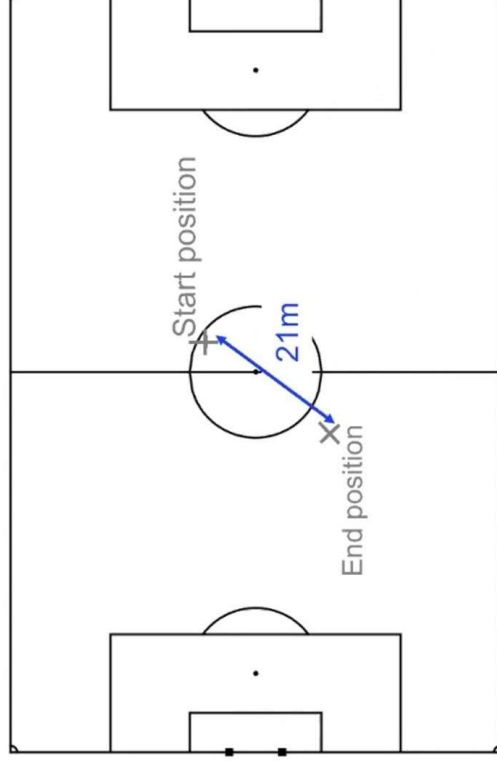
1. How long would it take for the ball to arrive?
2. How long would it take for each player to arrive?
3. What is the total probability that each team will control the ball after it has arrived?

Repeat calculations for all locations on pitch

Based on Laurie shaw lecture– [Advanced football analytics: building and applying a pitch control model in python.](#)

Building a Pitch Control Model

Ball Arrival Time



Assume ball speed = 15m/s
Arrival time = $21 / 15 = 1.4$ seconds

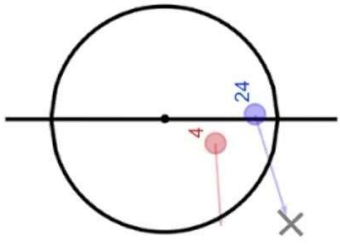
Building a Pitch Control Model

Player Arrival Time

Given their *initial position* and *velocity* how fast can a player get to any other position on the pitch?

Key assumptions:

- Players have a maximum speed: 5 m/s
- Players have a maximum acceleration: 7 m/s/s
- Players take the fastest possible path



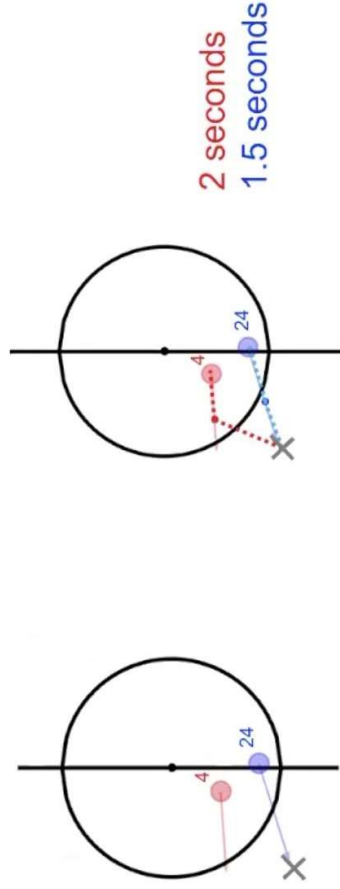
Not an easy problem to solve efficiently!

Building a Pitch Control Model

Simple approximation

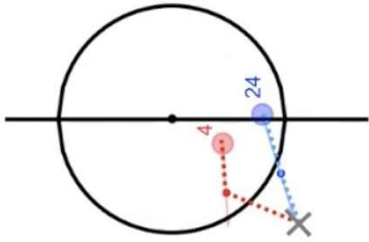
Two step process:

- There is an initial 'reaction time' of 0.7 seconds. During this time each player continues along their current trajectory.
- After 0.7 seconds, the player runs directly towards the target location at their maximum speed of 5 m/s.

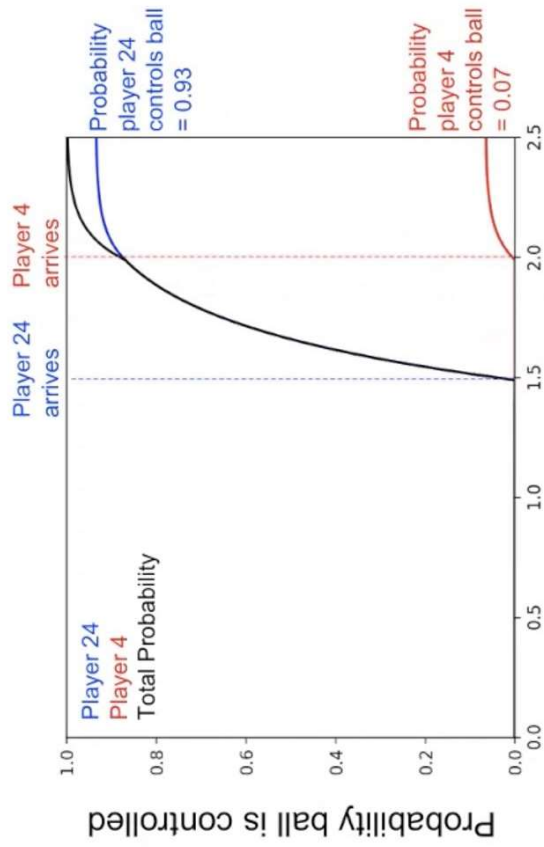


Building a Pitch Control Model

Exact Arrival Time Used

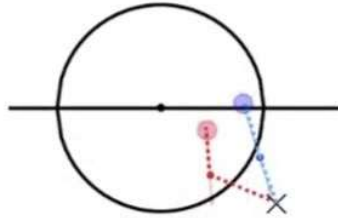


Ball arrives at 1.4 seconds
 Player 24 arrives after 1.5 seconds
 Player 4 arrives after 2 seconds

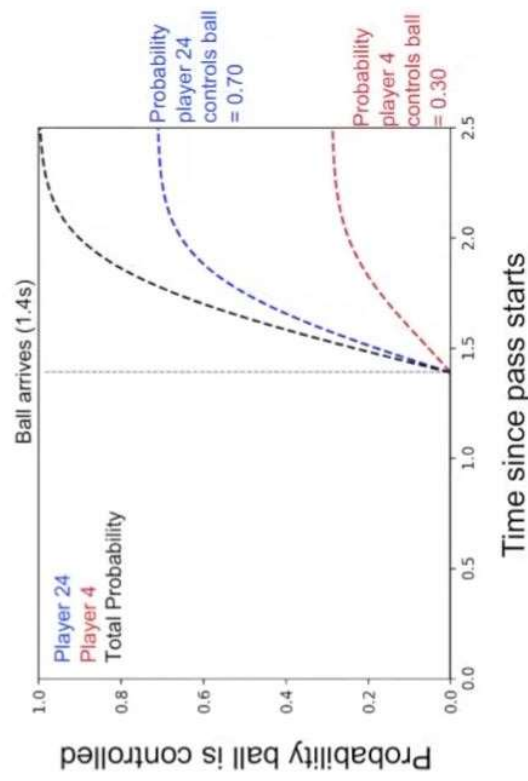


Building a Pitch Control Model

Uncertain Arrival Time Used

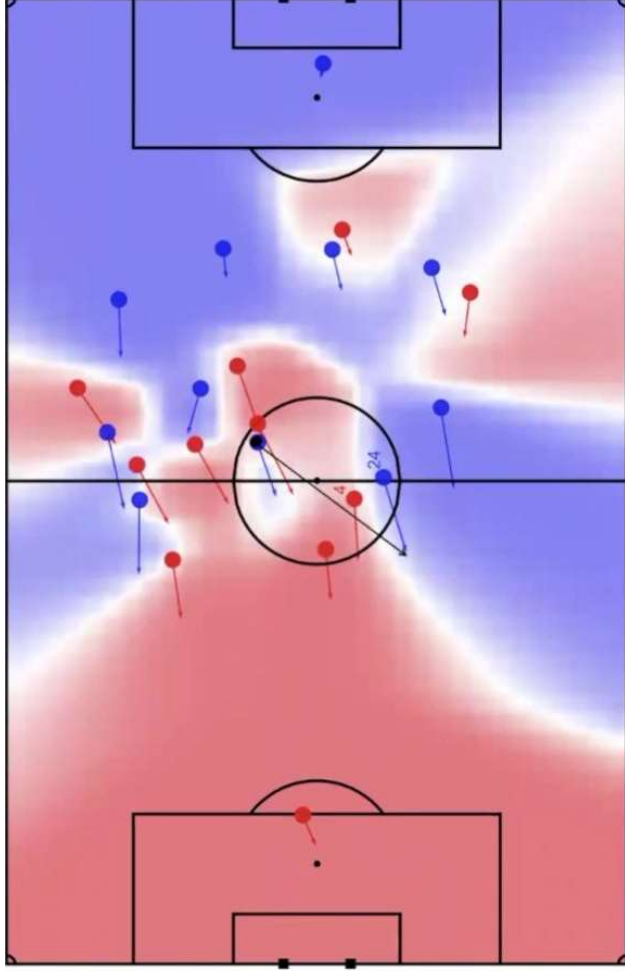


Ball arrives at 1.4 seconds
 Player 24 arrives after 1.5 seconds
 Player 4 arrives after 2 seconds



Building a Pitch Control Model

Final Product



Defence in Attack

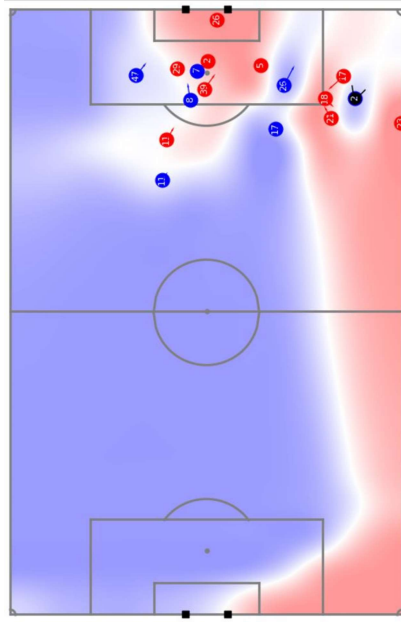


What is interesting about the position of the player on the bottom right of the 1st image?

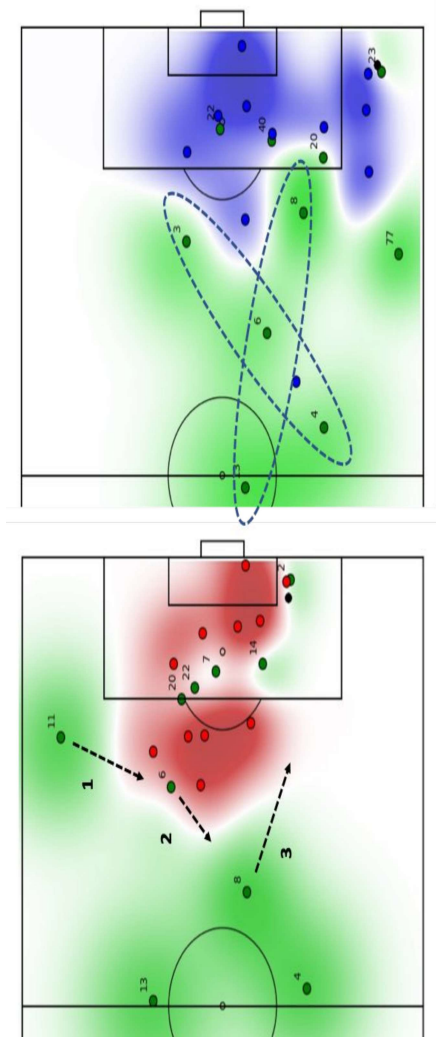
—His natural position is left back.

Why he is there? – Controlling dangerous zone while defending. Remember 5 second rule.

N:B: We want to open up dangerous zone while attacking.



Defence in Attack



Which one represents good positioning? - Right one!

- Dynamic solutions (formations) required for a wide variety of situations.
- A player's task is to find the empty point in the defensive shape (in this case a point in the cross) and occupy it. **No central planning is required**, just an understanding which positions need to be filled. Does it sound familiar?